A Corrected Note on W4385873460_p0: Partial Progress Toward ε-Equilibria in Stochastic Games with Borel Payoffs

Mathematical verification and revision

June 23, 2026

Abstract

Flesch and Solan ask whether every multiplayer stochastic game with finitely many players and actions, finite or countably infinite state space, and bounded Borel payoff functions admits an ε-Nash equilibrium for every $\varepsilon > 0$. This note is not a solution. It gives corrected and verified partial progress. First, the requirement that one profile work for all initial states is equivalent to the fixed-initial-state problem. Second, the desired conclusion holds for payoffs that are uniformly approximable by finite-prefix payoffs; in particular, it holds for finite-state games with continuous payoffs on the run space. Third, a compact-normal-form route to the theorem is impossible in this generality, as shown by a compact Borel zero-sum game with no small approximate saddle point. Fourth, the pure-threat lemma in the earlier draft is false as stated; it is replaced here by a correct first-deviation trigger theorem. Finally, a likelihood-ratio monitoring estimate is recorded, together with its limitation for arbitrary Borel payoffs.

Status. This note obtains partial progress only. It neither proves nor disproves the full Flesch–Solan open problem. The strongest positive theorem below is the elementary uniform finite-prefix approximation theorem. The strongest structural repair is the corrected pure-trigger criterion in Theorem 5.2. Both are genuine statements, but neither supplies the missing argument for arbitrary bounded Borel payoffs.

1 The model and the target problem

Let

$$\Gamma = (I, S, (A_i)_{i\in I}, p, (f_i)_{i\in I})$$

be a stochastic game. The player set I is finite, S is finite or countably infinite, each action set A_i is finite, $A = \prod_{i\in I} A_i$, and

$$p(\cdot \mid s, a) \in \Delta(S), \qquad s \in S,\ a \in A,$$

is the transition kernel. A run is

$$r = (s_1, a^1, s_2, a^2, \dots) \in S \times (A \times S)^{\mathbb{N}},$$

and each payoff f_i is bounded and Borel measurable on the run space.

For $n \geq 1$, let

$$H_n = S \times (A \times S)^{n-1}$$

be the set of histories before stage n, and let $H = \bigcup_{n\geq 1} H_n$. If $h = (s_1, a^1, \ldots, a^{n-1}, s_n) \in H_n$, write $s(h) = s_n$. A behavior strategy of player i assigns to each $h \in H$ a mixed action in $\Delta(A_i)$. A profile σ and an initial history h induce a probability measure $\mathbb{P}_{h,\sigma}$ and expectation operator $\mathbb{E}_{h,\sigma}$.

For $\varepsilon \geq 0$, a strategy profile σ^* is an ε-equilibrium from initial state s if for every player i and every unilateral deviation τ_i,

$$\mathbb{E}_{s,\sigma^*}[f_i] \geq \mathbb{E}_{s,(\tau_i,\sigma^*_{-i})}[f_i] - \varepsilon.$$

It is an all-state ε-equilibrium if this inequality holds for every initial state $s \in S$. Flesch and Solan prove several weaker existence results: subgame maxmin strategies, minmax acceptable profiles, extensive-form correlated ε-equilibria, and an ε-solvable subgame [1]. The missing object is an ordinary Nash ε-equilibrium for the original stochastic game.

2 The all-initial-states clause is harmless

The first observation is a useful reduction. It is not deep, but it prevents a false source of difficulty.

Lemma 2.1 (Pasting over initial states)**.** *Suppose that for every game of the above type, every initial state $s \in S$, and every $\varepsilon > 0$, there is an ε-equilibrium profile for the game started from s. Then every game of the above type has a single all-state ε-equilibrium profile.*

Proof. For each $s \in S$, choose an ε-equilibrium profile σ^s for the game started from s. Since every finite history records its initial state, define a single profile σ by

$$\sigma_i(h) = \sigma_i^{s_1}(h),$$

where s_1 is the first state in the history h. If the actual initial state is s, the play generated by σ is exactly the play generated by σ^s, and every unilateral deviation from that initial state is evaluated against σ^s_{-i}. Hence the ε-equilibrium inequalities for σ^s imply the corresponding inequalities for σ from s. Since s was arbitrary, σ works from all initial states. $\square$

Thus the fixed-initial-state problem is equivalent to the stated all-initial-states problem. The hard part is existence from one initial state.

3 A positive result for uniformly finite-prefix payoffs

This section records a clean subclass for which the desired conclusion follows by finite backward induction. The result is elementary, but it is a genuine positive statement and is useful as a benchmark for the Borel problem.

Definition 3.1 (Finite-prefix and uniformly finite-prefix approximable payoffs)**.** *A payoff g_i is an N-prefix payoff if it is measurable with respect to the first N stages, equivalently if it is a bounded function of*

$$(s_1, a^1, s_2, \ldots, a^N, s_{N+1}) \in H_{N+1}.$$

The payoff vector $f = (f_i)_{i\in I}$ is uniformly finite-prefix approximable if for every $\delta > 0$ there are N-prefix payoffs g_i such that

$$\sup_r |f_i(r) - g_i(r)| \leq \delta, \qquad \forall i \in I.$$

Lemma 3.2 (Finite-horizon stochastic games). *If every payoff g_i is an N-prefix payoff, then the stochastic game has a strategy profile that is an exact subgame-perfect equilibrium for the payoff vector g. In particular, it is an exact equilibrium from every initial state.*

Proof. We use backward induction over histories. For $h \in H_{N+1}$, set

$$V_i(h) = g_i(h), \qquad i \in I.$$

Assume V_i has been defined on H_{n+1}, where $1 \le n \le N$. For each $h \in H_n$, define a finite normal-form game $G(h)$ with player set I, action sets $(A_i)_{i \in I}$, and payoff to player i under action profile $a \in A$ equal to

$$G_i(h,a) = \sum_{s' \in S} p(s' \mid s(h), a)\, V_i(h, a, s').$$

The sum is well-defined because V_i is bounded and $p(\cdot \mid s(h), a)$ is a probability distribution on the countable set S. By Nash's theorem, the finite game $G(h)$ has a mixed equilibrium $x(h) \in \prod_i \Delta(A_i)$. Choose one such equilibrium for each $h \in H_n$ and define

$$V_i(h) = G_i(h, x(h)).$$

Proceeding backward to $n = 1$ defines $x(h)$ for every history before stage $N+1$. Let σ play $x(h)$ at such histories and play arbitrarily afterwards.

The usual backward-induction verification applies verbatim. At terminal histories no future action affects g. If the continuation profile is an equilibrium in every subgame after histories in H_{n+1}, then at a history $h \in H_n$ no player can gain by deviating at the current stage because $x(h)$ is a Nash equilibrium of $G(h)$; after the current stage the induction hypothesis applies. Hence σ is a subgame-perfect exact equilibrium for g. ☐

Theorem 3.3 (Uniform finite-prefix approximation gives ε-equilibria). *If f is uniformly finite-prefix approximable, then for every $\varepsilon > 0$ the stochastic game admits an all-state ε-equilibrium.*

Proof. Choose N-prefix payoffs g_i with

$$\sup_r |f_i(r) - g_i(r)| \le \varepsilon/2$$

for every player i. By Lemma 3.2, choose an exact equilibrium σ for the finite-prefix payoff vector g. For every initial state s, every player i, and every unilateral deviation τ_i,

$$\begin{aligned}
\mathbb{E}_{s,\sigma}[f_i] &\ge \mathbb{E}_{s,\sigma}[g_i] - \varepsilon/2 \\
&\ge \mathbb{E}_{s,(\tau_i,\sigma_{-i})}[g_i] - \varepsilon/2 \\
&\ge \mathbb{E}_{s,(\tau_i,\sigma_{-i})}[f_i] - \varepsilon.
\end{aligned}$$

Thus σ is an all-state ε-equilibrium for f. ☐

Corollary 3.4 (Finite state and continuous payoff). *If S is finite and each f_i is continuous on the run space $S \times (A \times S)^{\mathbb{N}}$ with the product topology, then the game admits an all-state ε-equilibrium for every $\varepsilon > 0$.*

Proof. When S and A are finite discrete spaces, the run space is compact metrizable. A continuous payoff is therefore uniformly continuous. Hence, for every $\delta > 0$, there is a prefix length N such that two runs with the same first N stages have payoff values within δ for every player. Choosing one representative continuation for each prefix gives N-prefix payoffs uniformly within δ of f_i. The conclusion follows from Theorem 3.3. ☐

Remark 3.5. *Theorem 3.3 does not reach arbitrary Borel payoffs. For example, on the binary run space $\{0,1\}^{\mathbb{N}}$, the indicator of the event "infinitely many ones occur" is Borel and tail measurable, but no finite-prefix payoff can approximate it uniformly within error smaller than $1/2$: two runs can share any prescribed finite prefix while one has infinitely many ones and the other is eventually all zero. Thus finite-prefix approximation is a useful positive region, not a solution of the Borel problem.*

4 Why compact strategic-form Borel equilibrium is not enough

For a fixed initial state, the pure strategy space of each player is a countable product of finite sets, hence compact metrizable. It is tempting to hope for a general theorem asserting that compact games with bounded Borel payoffs have countably additive mixed ε-equilibria. Such a theorem is false even in zero-sum games.

Proposition 4.1 (Compact Borel Wald game)**.** *Let $X = Y = \mathbb{N} \cup \{\infty\}$ with the one-point compactification topology. Define a zero-sum payoff $u : X \times Y \to \{0,1\}$ by*

$$u(x,y) = \begin{cases} 1, & x, y \in \mathbb{N} \text{ and } x \geq y, \\ 1, & x \in \mathbb{N},\ y = \infty, \\ 0, & x = \infty. \end{cases}$$

Then X and Y are compact metrizable, u is Borel, and the game has no countably additive mixed ε-equilibrium for any $\varepsilon < 1/2$.

Proof. The set $u^{-1}(\{1\})$ is

$$\{(x,y) \in \mathbb{N}^2 : x \geq y\} \cup (\mathbb{N} \times \{\infty\}),$$

a countable union of Borel rectangles. Hence u is Borel.

Let $U(\mu,\nu) = \int u \, d\mu d\nu$. Fix a countably additive probability μ on X. For finite y,

$$U(\mu, \delta_y) = \mu(\{x \in \mathbb{N} : x \geq y\}) \longrightarrow 0$$

as $y \to \infty$. Hence $\inf_\nu U(\mu,\nu) = 0$.

Conversely, fix a countably additive probability ν on Y. For finite x,

$$U(\delta_x, \nu) = \nu(\{y \in \mathbb{N} : y \leq x\}) + \nu(\{\infty\}) \longrightarrow 1$$

as $x \to \infty$. Hence $\sup_\mu U(\mu,\nu) = 1$.

If (μ,ν) were a zero-sum mixed ε-equilibrium, player 2's optimality would imply $U(\mu,\nu) \leq \varepsilon$, while player 1's optimality would imply $U(\mu,\nu) \geq 1 - \varepsilon$. This is impossible when $\varepsilon < 1/2$. □

Remark 4.2. *Proposition 4.1 is not a stochastic-game counterexample. Its role is more limited: it shows that a proof of the open problem must exploit the causal and public monitoring structure of stochastic games, rather than merely compactness of pure strategy spaces and Borel measurability of payoffs.*

5 Correcting the pure-threat argument

The earlier draft contained a pure-threat lemma asserting, in essence, that pure subgame maxmin strategies plus pure punishments imply an approximate equilibrium when deviations are observed. The following one-stage game shows that this is not correct for simultaneous games.

Proposition 5.1 (Failure of the naive pure-threat lemma). *There is a finite one-stage stochastic game in which both players have pure maxmin actions and pure punishment actions, but there is no pure ε-equilibrium for any $\varepsilon < 1$.*

Proof. Consider the two-player one-stage game with actions T, B for player 1 and L, R for player 2. After the action profile is chosen, the game is absorbed and the payoffs are

$$u_1 = \begin{array}{c|cc} & L & R \\ \hline T & -1 & 1 \\ B & 0 & 0 \end{array} \qquad u_2 = \begin{array}{c|cc} & L & R \\ \hline T & 0 & -1 \\ B & 0 & 1 \end{array}.$$

For player 1, the maxmin value is 0: action B guarantees 0, and player 2 can hold player 1 to at most 0 by playing L. For player 2, the maxmin value is also 0: action L guarantees 0, and player 1 can hold player 2 to at most 0 by playing T. Thus both players have pure maxmin strategies and pure punishments.

Nevertheless, no pure profile is an ε-equilibrium for $\varepsilon < 1$:

profile	profitable pure deviation
(T, L)	player 1 switches to B and gains 1,
(T, R)	player 2 switches to L and gains 1,
(B, L)	player 2 switches to R and gains 1,
(B, R)	player 1 switches to T and gains 1.

The original pure-threat claim would incorrectly predict a pure approximate equilibrium. The reason is visible at (B, L): if player 2 deviates to R, the play moves to a terminal history where player 2's payoff is already 1, so punishment after the detected action cannot undo the favorable jump. $\square$

A correct pure trigger statement must control the payoff after the first detected off-path action. The next theorem gives such a criterion.

Fix a pure strategy profile q. For each player i and history h, write

$$G_i^q(h) := \mathbb{E}_{h,q}[f_i]$$

for player i's continuation payoff if the pure profile q is followed forever from h. For each player i and each one-step successor history $h^+ = (h, a, s')$, fix an opponents' strategy profile π^{-i,h^+} in the subgame at h^+. If, at history h, player i chooses $b_i \in A_i$ while all other players choose the pure actions $q_{-i}(h)$, define

$$D_i(h, b_i) := \sum_{s' \in S} p(s' \mid s(h), (b_i, q_{-i}(h))) \sup_{\tau_i} \mathbb{E}_{h^+(s'),(\tau_i,\pi^{-i,h^+(s')})}[f_i],$$

where

$$h^+(s') = (h, (b_i, q_{-i}(h)), s').$$

Thus $D_i(h, b_i)$ is the best expected payoff that player i can secure after the first detected off-path action b_i at h, assuming the opponents punish from the next history onward.

Theorem 5.2 (First-deviation trigger criterion). *Assume that there is $\eta \geq 0$ such that, for every player i, every history h, and every action $b_i \neq q_i(h)$,*

$$D_i(h, b_i) \leq G_i^q(h) + \eta. \tag{FD}$$

Construct a strategy profile σ^* as follows: play q until the first observed deviation from q; if player i is the first deviator and the next history is h^+, then all opponents of i switch to π^{-i,h^+} forever. Then σ^* is an all-state η-equilibrium.

Proof. Fix an initial state s, a player i, and a unilateral deviation τ_i. Work on a reference probability space on which the public actions follow the pure profile q from s, and adjoin the private randomization used by τ_i. Let K_n be the public history before stage $n+1$ in this reference play, so $K_n \in H_{n+1}$ and $K_0 = s$. Let $\mathcal{F}_n$ be the sigma-field generated by K_n; equivalently, by the public history up to that point. Define the first-deviation epoch

$$\rho = \inf\{n \geq 0 : \text{when fed the history } K_n,\ \tau_i \text{ chooses an action } b_i \neq q_i(K_n)\},$$

with $\rho = \infty$ if the set is empty. On $\{\rho < \infty\}$, write B_ρ for the first off-path action. Up to epoch ρ, the public history in the actual play against the trigger profile agrees with the reference history K_n, because before ρ the realized action of player i is exactly the prescribed action q_i.

Under q, the process

$$M_n = \mathbb{E}_{s,q}[f_i \mid \mathcal{F}_n]$$

is a bounded martingale. Since K_n is the whole public history before stage $n+1$,

$$M_n = G_i^q(K_n).$$

Moreover, because f_i is Borel measurable with respect to the infinite public run, Levy's martingale convergence theorem gives $M_n \to f_i$ almost surely and in L^1. Optional sampling applied to the bounded stopping times $n \wedge \rho$, followed by bounded convergence, yields

$$\mathbb{E}\big[G_i^q(K_\rho)\mathbf{1}_{\{\rho<\infty\}} + f_i\mathbf{1}_{\{\rho=\infty\}}\big] = G_i^q(s). \tag{1}$$

On $\{\rho = \infty\}$, the deviating player has never chosen an action different from q_i, so the realized run is the same as under q. On $\{\rho < \infty\}$, conditional on $K_\rho = h$ and $B_\rho = b_i$, the expected payoff against the trigger profile is at most $D_i(h, b_i)$, because after the next state is drawn the opponents use the specified punishment profile and the deviator is allowed an arbitrary continuation strategy in the supremum defining $D_i(h, b_i)$. By condition (FD), this conditional payoff is at most $G_i^q(h) + \eta$. Hence, using (1),

$$\begin{aligned}
\mathbb{E}_{s,(\tau_i,\sigma^*_{-i})}[f_i] &\leq \mathbb{E}\big[(G_i^q(K_\rho) + \eta)\mathbf{1}_{\{\rho<\infty\}} + f_i\mathbf{1}_{\{\rho=\infty\}}\big] \\
&\leq G_i^q(s) + \eta.
\end{aligned}$$

Under σ^* with no deviation, the profile follows q forever, so $\mathbb{E}_{s,\sigma^*}[f_i] = G_i^q(s)$. Therefore player i can gain at most η. The initial state, player, and deviation were arbitrary. $\square$

The criterion can be expressed in terms of punishment values.

Corollary 5.3 (No upward punishment-value jump)**.** *For each player i and history h, define*

$$m_i(h) = \inf_{\pi_{-i}} \sup_{\tau_i} \mathbb{E}_{h,(\tau_i,\pi_{-i})}[f_i].$$

Suppose that, for some $\delta, \eta \geq 0$, the chosen punishment profiles satisfy

$$\sup_{\tau_i} \mathbb{E}_{h,(\tau_i,\pi^{-i,h})}[f_i] \leq m_i(h) + \delta \qquad \textit{for every } i, h,$$

and that the pure reference profile q satisfies, for every i, h and every $b_i \neq q_i(h)$,

$$\sum_{s' \in S} p(s' \mid s(h), (b_i, q_{-i}(h)))m_i\big(h, (b_i, q_{-i}(h)), s'\big) \leq G_i^q(h) + \eta - \delta. \tag{NJ}$$

Then the trigger profile associated with q and these punishments is an all-state η-equilibrium.

Proof. The punishment inequality implies

$$D_i(h, b_i) \le \sum_{s' \in S} p(s' \mid s(h), (b_i, q_{-i}(h)))\big[m_i(h, (b_i, q_{-i}(h)), s') + \delta\big].$$

Condition (NJ) gives $D_i(h, b_i) \le G_i^q(h) + \eta$, so Theorem 5.2 applies. $\square$

Remark 5.4. *Proposition 5.1 violates condition* (NJ)*. At the profile* (B, L)*, player 2's continuation payoff is* 0*, but the one-stage deviation to* R *creates an absorbing continuation with payoff* 1*. This is exactly the upward jump that the earlier proof overlooked.*

6 A likelihood-ratio monitoring estimate and its limitation

The following estimate is correct and often useful, but it is too weak by itself to prove the open problem.

Lemma 6.1 (Likelihood-ratio alarm bound)**.** *Let* $(\Omega, \mathcal{F}_\infty)$ *be a measurable space with filtration* $(\mathcal{F}_n)_{n \ge 0}$ *such that* $\mathcal{F}_\infty = \sigma(\bigcup_n \mathcal{F}_n)$*. Let* P *and* Q *be probability measures such that* $Q|_{\mathcal{F}_n} \ll P|_{\mathcal{F}_n}$ *for every* n*, and let*

$$L_n = \frac{dQ|_{\mathcal{F}_n}}{dP|_{\mathcal{F}_n}}.$$

For $T > 1$*, set* $\theta_T = \inf\{n : L_n \ge T\}$*. Then*

$$P(\theta_T < \infty) \le \frac{1}{T}.$$

Moreover, for every $B \in \mathcal{F}_\infty$*,*

$$Q(B \cap \{\theta_T = \infty\}) \le T P(B).$$

Proof. The process (L_n) is a nonnegative P-martingale with $\mathbb{E}_P L_n = 1$. Doob's maximal inequality gives

$$P(\theta_T < \infty) = P\left(\sup_n L_n \ge T\right) \le \frac{1}{T}.$$

For the domination claim, first take $B \in \mathcal{F}_m$. Then

$$\begin{aligned} Q(B \cap \{\theta_T = \infty\}) &\le Q(B \cap \{\theta_T > m\}) \\ &= \mathbb{E}_P[\mathbf{1}_B \mathbf{1}_{\{\theta_T > m\}} L_m] \le T P(B), \end{aligned}$$

because $L_m < T$ on $\{\theta_T > m\}$. The class of $B \in \mathcal{F}_\infty$ for which the inequality holds is a monotone class containing the pi-system $\bigcup_m \mathcal{F}_m$, and therefore contains $\mathcal{F}_\infty$. $\square$

Remark 6.2. *In a stochastic game, if a prescribed behavior profile has full support, then every unilateral deviation is absolutely continuous with respect to it on each finite-stage sigma-field, so the lemma applies. It controls false alarms under the prescribed law and dominates the deviating law restricted to the no-alarm event. It still does not yield a Nash equilibrium for arbitrary Borel payoffs: such payoffs can be sensitive to events whose probabilities are changed substantially by a deviation, and the no-alarm domination constant* T *must be large when the false-alarm probability* $1/T$ *is small.*

7 Conclusion

The corrected status is as follows.

1. This note is related to the Flesch–Solan problem, but it does not solve it.
2. The all-initial-states formulation is equivalent to the fixed-initial-state problem by Lemma 2.1.
3. The problem has an affirmative answer for uniformly finite-prefix approximable payoffs, including finite-state continuous payoffs, by Theorem 3.3 and Corollary 3.4.
4. A compact Borel strategic-form theorem cannot solve the problem, because such a theorem is false even for compact zero-sum games (Proposition 4.1).
5. The earlier pure-threat lemma was mathematically false. The corrected theorem requires the first-deviation condition (FD), or sufficient no-jump condition (NJ), and Proposition 5.1 shows why such a condition cannot be omitted.
6. Likelihood-ratio detection gives rigorous false-alarm control, but it does not control arbitrary Borel payoff changes caused by statistically subtle deviations.

Accordingly, the result obtained here should be described as partial progress and error-correction, not as a solution. The remaining hard step is to bridge the gap between Flesch–Solan minmax acceptable profiles and ordinary Nash ε-equilibria without a mediator, in the presence of simultaneous mixed actions and arbitrary bounded Borel payoffs.

References

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